

Introduction to Mathematics and Optimization

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PROBLEM 1

Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be the function given by

$$f(x, y, z) = x^2 + y^2$$

and let $C \subseteq \mathbb{R}^3$ be the subset

$$C = \{(x, y, z) \in \mathbb{R}^3 \mid y + z \geq 1, x + z \geq 1, 2x + y + z \leq -1, x + 2y + z \leq -1\}.$$

We consider the optimization problem

$$\begin{aligned} \min f(x, y, z) \\ (x, y, z) \in C. \end{aligned} \tag{1}$$

- (a) Show that f is a convex function, but not a strictly convex function.
- (b) Use Fourier-Motzkin elimination of z to show that $(x, y, z) \in C$ implies $x \leq -1$ and $y \leq -1$. Can one conclude from this that C is not bounded?
- (c) Show that (1) is a convex optimization problem.
- (d) Write down the KKT conditions for the optimization problem (1). Use only references to and concepts from the syllabus.
- (e) Use the KKT conditions to show that $(-2, -2, 4)$ is not a solution to the optimization problem (1).
- (f) It is known that $(-1, -1, 2)$ is a solution to the optimization problem (1). Are there more than one solution to the KKT conditions for this optimal solution?

PROBLEM 2

For $a \in \mathbb{R}$ the matrix M_a is defined by

$$M_a = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & a \end{pmatrix}.$$

For example

$$M_0 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

- (a) Use an algorithm from the syllabus to show that M_1 is positive semidefinite.
- (b) Find the set of all $a \in \mathbb{R}$ such that M_a is positive definite, again using the algorithm from the syllabus.
- (c) Is M_1 invertible?
- (d) Find a function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ such that M_2 is the Hessian matrix of f . Is this function strictly convex?

PROBLEM 3

Consider the function $f_a : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by

$$f_a(x, y) = x^2 + axy + y^2$$

where $a \in \mathbb{R}$ is a real constant. For example $f_2(x, y) = x^2 + 2xy + y^2$. Also let

$$C = \{(x, y) \in \mathbb{R}^2 \mid x \leq 0, y \leq 0, x + y \geq -2\}.$$

- (a) Show that C is bounded using the definition in the syllabus.
- (b) Investigate for which values of a the function f_a is convex.
- (c) Find all solutions to the optimization problem

$$\begin{aligned} \min f_{-2}(x, y) \\ (x, y) \in C. \end{aligned}$$